Chapter 4 — Stoichiometry

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Mole concept and molar mass

Q.1 Explain the mole concept and derive relationships between mass, moles, molar mass and Avogadro's number.

A mole is the amount of substance containing as many elementary entities as there are atoms in 12 g of carbon-12. This number is Avogadro's number, $N_A = 6.02 \times 10^{23}$ mol^{-1}. The entities may be atoms, molecules, ions or formula units.

Molar mass, M, is the mass of one mole in grams. The basic relationships are: $n = m/M$, $m = nM$ and number of particles $= nN_A$. For example, 20 g NaOH is 20/40 = 0.5 mol, and 0.5 mol HCl has mass $0.5 \times 36.5 = 18.25$ g.

Textbook check (p. 72): Sample Problem 4.2 prints the mass of HCl as "18.3 mol"; the correct unit is grams, about 18.3 g.

Important Conceptual Short Questions — Topic 1

1. Why is the mole concept necessary for chemical calculations?

Atoms and molecules are too small to count individually. The mole links the microscopic number of particles to a measurable laboratory mass.

2. How does molar mass connect atomic mass with laboratory mass?

The atomic or molecular mass in atomic mass units becomes the molar mass in grams per mole. Thus one mole of a substance has a mass in grams numerically equal to its relative mass.

3. Why do equal moles of different gases contain the same number of molecules?

A mole always contains Avogadro's number of entities. Therefore equal moles contain equal numbers of molecules, although their masses may differ.

4. How can the number of atoms in a given mass of an element be calculated?

First find moles using $n = m/M$, then multiply by Avogadro's number. For a molecular element, also multiply by the number of atoms per molecule if atoms are required.

Topic 2 — Molar volume, gas density and molar concentration

Q.2 Explain molar volume, molar mass from gas density, and molar concentration.

Molar volume is the volume of one mole of an ideal gas at STP. The textbook uses STP as 273.16 K and 1 atm, where molar volume is 22.414 dm^3, commonly rounded to 22.4 dm^3. Therefore $V = n \times 22.4$ at STP.

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Gas density can be used to find molar mass: M = density × molar volume. Molar concentration is moles of solute per dm^3 of solution: c = n/V. For example, 4.0 g NaOH in 250 cm^3 gives 0.10 mol in 0.250 dm^3, so c = 0.40 mol dm^{-3}.

Important Conceptual Short Questions — Topic 2

1. Why does one mole of different ideal gases occupy the same volume at STP?

Avogadro's law states that equal volumes of gases at the same temperature and pressure contain equal numbers of molecules. One mole of any ideal gas therefore occupies the same molar volume.

2. How can gas density be used to determine molar mass?

At STP, one mole occupies 22.4 dm^3. Multiplying the density in g dm^{-3} by 22.4 gives the mass of one mole, which is the molar mass.

3. Why is molar concentration more useful than mass concentration in reactions?

Chemical equations react in mole ratios, not mass ratios. Molar concentration directly gives the number of reacting particles in a measured volume.

4. How do temperature and pressure affect the volume of a gas?

At constant pressure, increasing temperature increases volume. At constant temperature, increasing pressure decreases volume. Therefore gas volumes must be compared at the same temperature and pressure.

Topic 3 — Stoichiometric calculations and percentage yield

Q.3 Describe the approach to stoichiometric calculations and explain theoretical, actual and percentage yield.

A balanced chemical equation gives the mole ratio of reactants and products. The usual method is: convert the given quantity to moles, use the mole ratio, then convert to the required mass, volume or number of particles. For gases at STP, 1 mol = 22.4 dm^3; for solutions, n = cV.

Theoretical yield is the maximum amount calculated from the balanced equation. Actual yield is the amount obtained in the laboratory. Percentage yield = (actual yield/theoretical yield) × 100. Actual yield is usually lower because of side reactions, incomplete reaction, reversible reactions and losses during separation and purification.

Important Conceptual Short Questions — Topic 3

1. Why must a chemical equation be balanced before stoichiometric calculation?

Balancing ensures that atoms are conserved. The coefficients then give the correct mole ratios needed for calculation.

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2. How does the mole ratio from a balanced equation determine product amount?

The coefficients show how many moles of each substance react or form. Once the moles of one substance are known, the mole ratio gives the moles of any other substance.

3. Why is actual yield usually less than theoretical yield?

Some reactant may remain unreacted, side products may form, the reaction may be reversible, or product may be lost during filtration, washing, drying or transfer.

4. How can percentage yield be used to judge the efficiency of a process?

A high percentage yield means that most of the theoretical product was obtained. It helps chemists compare methods and reduce waste.

Topic 4 — Limiting reactant and importance in medicine

Q.4 Explain limiting and excess reactants and the importance of stoichiometry in medicine.

The limiting reactant is the reactant that is completely consumed first and therefore controls the maximum amount of product. The excess reactant remains after the reaction. To identify it, calculate the moles of each reactant and compare them with the mole ratio in the balanced equation.

Stoichiometry is essential in medicine because the correct amount of active ingredient is needed for safety and effectiveness. It is used in drug synthesis, dosage preparation, blood glucose and cholesterol measurement, vaccine antigen concentration and quality control. Too little may be ineffective, while too much may be toxic.

Textbook check (p. 86): Sample Problem 4.20 contains printing slips: 3.00 g is written as 300 g in one step, 0.022 mol is later written as 0.021 mol, and the actual yield is printed as 2.85 g although the question states 3.15 g. Use the data given in the question and carry units consistently.

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Important Conceptual Short Questions — Topic 4

1. Why does the limiting reactant control the amount of product formed?

Once the limiting reactant is used up, the reaction cannot continue, even if other reactants remain. Therefore it fixes the maximum product amount.

2. How can the excess reactant remaining after a reaction be calculated?

Use the limiting reactant to calculate how much excess reactant was consumed. Subtract this from the initial amount of excess reactant.

3. Why is stoichiometry important in pharmaceutical manufacturing?

It ensures that reactants combine in the correct proportions, giving the intended drug with minimum impurities and a reliable dose.

4. How can incorrect stoichiometric dosage affect patient safety?

An underdose may fail to treat the disease, while an overdose may cause toxicity. Accurate stoichiometry keeps the active ingredient within the safe therapeutic range.

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